

UNIT – I

Introduction to Cellular Mobile Systems

1.1 Why cellular mobile communication systems?

LIMITATIONS OF CONVENTIONAL MOBILE TELEPHONE SYSTEMS

One of many reasons for developing a cellular mobile telephone system and deploying it in many cities is the operational limitations of conventional mobile telephone systems: limited service capability, poor service performance, and inefficient frequency spectrum utilization.

LIMITED SERVICE CAPABILITY:

A conventional mobile telephone system is usually designed by selecting one or more channels from a specific frequency allocation for use in autonomous geographic zones, as shown in Fig.1.1. The communications coverage area of each zone is normally planned to be as large as possible, which means that the transmitted power should be as high as the federal specification allows. The user who starts a call in one zone has to reinitiate the call when moving into a new zone because the call will be dropped. This is an undesirable radio telephone system since there is no guarantee that a call can be completed without a handoff capability.

The handoff is a process of automatically changing frequencies as the mobile unit moves into a different frequency zone so that the conversation can be continued in a new frequency zone without redialing. Another disadvantage of the conventional system is that the number of active users is limited to the number of channels assigned to a particular frequency zone.

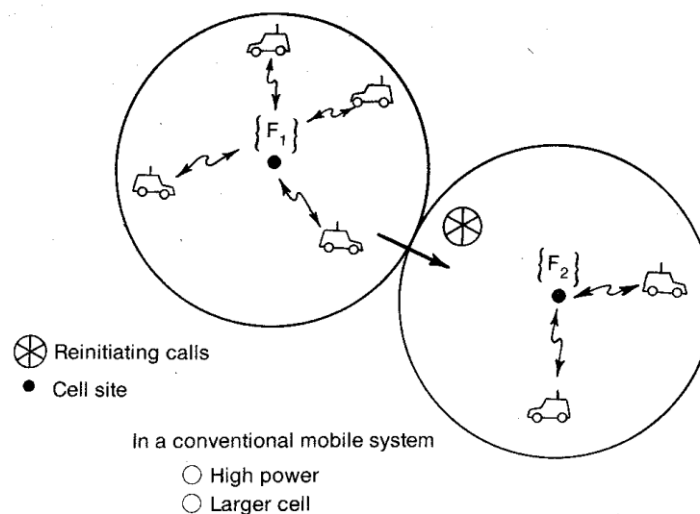


Figure 1.1 Conventional mobile system.

POOR SERVICE PERFORMANCE:

In the past, a total of 33 channels were all allocated to three mobile telephone systems: Mobile Telephone Service (MTS), Improved Mobile Telephone Service (IMTS) MJ systems and Improved Mobile Telephone Service (IMTS) MK systems. MTS operates around 40 MHz and MJ operates at 150 MHz; both provide 11 channels; IMTS MK operates at 450 MHz and provides 12 channels. These 33 channels must cover an area 50 mi in diameter. In 1976, New York City had 6 channels of MJ serving 320 customers, with another 2400 customers on a waiting list. New York City also had 6 channels of MK serving 225 customers, with another 1300 customers on a waiting list. The large number of subscribers created a high blocking probability during busy hours. Although service performance was undesirable, the demand was still great. A high-capacity system for mobile telephones was needed.

INEFFICIENT FREQUENCY SPECTRUM UTILIZATION:

In a conventional mobile telephone system, the frequency utilization measurement M_o , is defined as the maximum number of customers that could be served by one channel at the busy hour.

1.2 BASIC CELLULAR SYSTEM

A basic analog cellular system consists of three subsystems: a mobile unit, a cell site, and a mobile telephone switching office (MTSO), as Fig. 1.1 shows, with connections to link the three subsystems.

Mobile units: A mobile telephone unit contains a control unit, a transceiver, and an antenna system.

Cell site: The cell site provides interface between the MTSO and the mobile units. It has a control unit, radio cabinets, antennas, a power plant, and data terminals.

MTSO: The switching office, the central coordinating element for all cell sites, contains the cellular processor and cellular switch. It interfaces with telephone company zone offices, controls call processing, provides operation and maintenance, and handles billing activities.

Connections: The radio and high-speed data links connect the three subsystems. Each mobile unit can only use one channel at a time for its communication link. But the channel is not fixed; it can be any one in the entire band assigned by the serving area, with each site having multichannel capabilities that can connect simultaneously to many mobile units.

The MTSO is the heart of the analog cellular mobile system. Its processor provides central coordination and cellular administration. The cellular switch, which can be either analog or digital, switches calls to connect mobile subscribers to other mobile subscribers and to the nationwide telephone network. It uses voice trunks similar to telephone company interoffice voice trunks. It also contains data links providing supervision links between the processor and the switch and between the cell sites and the processor. The radio link carries the voice and signaling between the mobile unit and the cell site. The high-speed data links cannot be transmitted over the standard telephone trunks and therefore must use either microwave links or T-carriers (wire lines). Microwave radio links or T-carriers carry both voice and data between cell site and the MTSO.

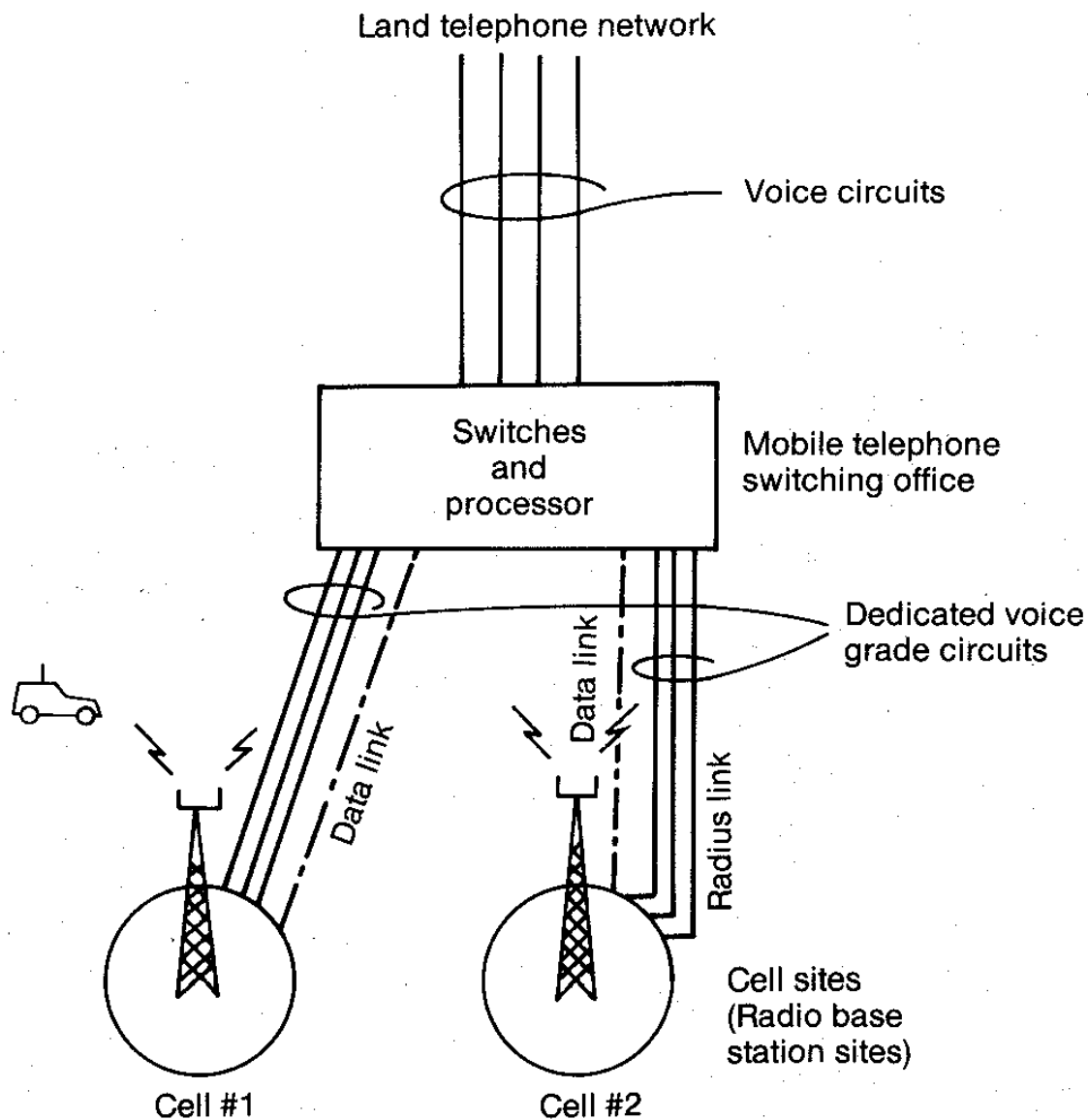


Figure 1.3 Cellular system.

1.3 EVOLUTION OF MOBILE RADIO COMMUNICATIONS

The first wireline telephone system was introduced in the year 1877. Mobile communication systems as early as 1934 were based on Amplitude Modulation (AM) schemes and only certain public organizations maintained such systems. With the demand for newer and better mobile radio communication systems during the World War II and the development of Frequency Modulation (FM) technique by Edwin Armstrong, the mobile radio communication systems began to witness many new changes. Mobile telephone was introduced in the year 1946. However, during its initial three and a half decades it found very less market penetration owing to high costs and

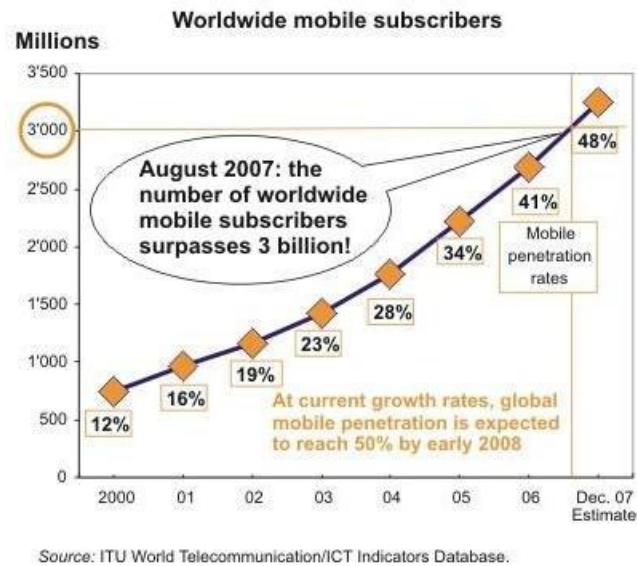


Figure 1.3: The worldwide mobile subscriber chart.

numerous technological drawbacks. But with the development of the cellular concept in the 1960s at the Bell Laboratories, mobile communications began to be a promising field of expanse which could serve wider populations. Initially, mobile communication was restricted to certain official users and the cellular concept was never even dreamt of being made commercially available. Moreover, even the growth in the cellular networks was very slow. However, with the development of newer and better technologies starting from the 1970s and with the mobile users now connected to the Public Switched Telephone Network (PSTN), there has been an astronomical growth in the cellular radio and the personal communication systems. Advanced Mobile Phone System (AMPS) was the first U.S. cellular telephone system and it was deployed in 1983. Wireless services have since then been experiencing a 50% per year growth rate. The number of cellular telephone users grew from 25000 in 1984 to around 3 billion in the year 2007 and the demand rate is increasing day by day. A schematic of the subscribers is shown in Fig. 1.3.

Present Day Mobile Communication

Since the time of wireless telegraphy, radio communication has been used extensively. Our society has been looking for acquiring mobility in communication since then. Initially the mobile communication was limited between one pair of users on single channel pair. The range of mobility was defined by the transmitter power, type of antenna used and the frequency of operation. With the increase in the number of users, accommodating them within the limited available frequency spectrum became a major problem. To resolve this problem, the concept of cellular communication was evolved. The present day cellular communication uses a

basic unit called cell. Each cell consists of small hexagonal area with a base station located at the center of the cell which communicates with the user. To accommodate multiple users Time Division multiple Access (TDMA), Code Division Multiple Access (CDMA), Frequency Division Multiple Access (FDMA) and their hybrids are used. Numerous mobile radio standards have been deployed at various places such as AMPS, PACS, GSM, NTT, PHS and IS-95, each utilizing different set of frequencies and allocating different number of users and channels.

1.2 PERFORMANCE CRITERIA

Performance of cellular mobile communication is evaluated on following parameters:

➤ Voice Quality:

Voice quality is difficult to judge without subjective/practical tests for users' feedback. Voice quality is a major factor that is considered in the performance criteria of a cellular system. It is measured on the basis of circuit merits (CM). The voice quality will be based upon the following criterion: a set value x at which y percent of customers rate the system voice quality as a receiver as good or excellent, the top two circuit merits (CM) of the five listed below.

Circuit Merits (CM)	Score	Quality Scale
CM5	5	Excellent i.e. voice perfectly understandable
CM4	4	Good i.e. voice easily understandable but some noise is present
CM3	3	Fair i.e. voice understandable but required effort
CM2	2	Poor i.e. voice is understandable but required frequent repetition
CM1	1	Unsatisfactory i.e. voice not understandable

Service Quality

The service quality of cellular system is based on following parameters:

- Coverage: The coverage area of system should as large as possible. Practically it is not possible to cover 100 percent of the area because of irregular terrain configurations.
- Required grade of service: The blocking probability of each cell site will be different. To decrease the blocking probability of system required a good system plan and sufficient number of radio channels.
- Number of dropped calls: Number of call drop should be less as

possible. Suppose in an hour Q calls are made and among Q calls if a call is dropped. Then $Q-1$ calls will be completed then the call drop rate is given by $1/Q$. This drop rate must be kept low. A high drop rate occurs due to either by coverage problems or handoff problems

➤ **Special Features**

A system has to provide as many as special features such as the following

- Call Forwarding
- Call Waiting
- Voice Stored (VSR) Box
- Automatic Roaming
- Navigation Services.

1.5 CHARACTERISTICS OF MOBILE RADIO ENVIRONMENT (or)

UNIQUENESS OF MOBILE RADIO ENVIRONMENT

The uniqueness of mobile radio environment can be described on the basis of following points:

- Description of mobile radio transmission medium
- Model of transmission medium
- Direct wave path, line of sight path and obstructive path
- Noise level in cellular frequency band
- Amplifier noise.
- Mobile fading characteristics

➤ **Propagation Attenuation:**

- There are number of factors which contribute to the signal strength at receiver. For a line of sight (LOS) path, there is free space path loss which is related to the radial distance between transmitter and receiver.
- Propagation in free space and without any obstacle is the most ideal situation. Where there is no line of sight path or when radio wave reaches to an obstacle, various propagation effects like reflection, scattering, diffraction, refraction occurs.
- The propagation path loss increases with frequency and distance.

$$C \propto R^{-4}$$

$$C = \alpha R^{-4}$$

Where C is the received carrier power, α is constant and R is the distance between transmitter and receiver. The propagation path loss varies on the basis of environmental conditions.

➤ Fading

In wireless communications, fading is deviation of the attenuation affecting a signal over certain propagation media. The fading may vary with time, geographical position or radio frequency, and is often modeled as a random process. In wireless systems fading may either be due to multipath propagation, referred to as multipath induced fading, or due to shadowing from obstacles affecting the wave propagation, sometimes "referred to as shadow fading. The fading can be broadly classified into two categories:

1. Large scale fading
2. Small scale fading.

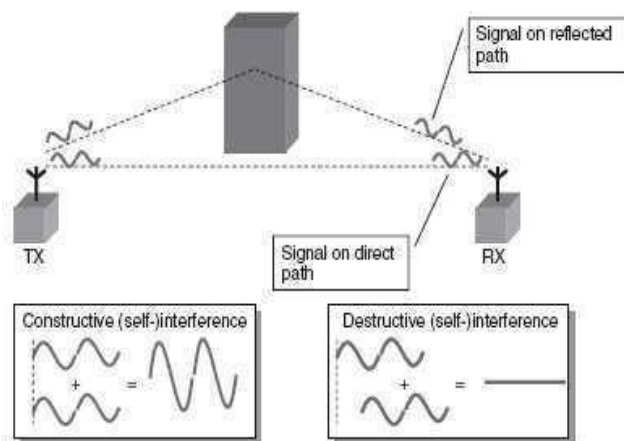


Figure 1.2 Principle of small-scale fading

• Types of fading Small Scale Fading

Small scale fading refers to rapid variation of instantaneous received signal strength due to the constructive and destructive interference of multi-paths when the mobile station moves over very short distances. Depending on the multipath delay spread small scale fading can be classified as either frequency-selective fading or frequency flat fading. Depending on the Doppler spread, small scale fading can be classified as either fast fading or slow fading.

Large Scale Fading

Large-scale fading occurs as the mobile moves over a large distance. It is caused by path loss of signal as a function of distance and shadowing by large objects such as intervening terrains, buildings, and vegetation. Shadowing is a slow fading process characterized by variation of median path loss between the transmitter and receiver in

fixed locations. The received signal strength may be different for even at the same distance from a transmitter, due to the shadowing caused by obstacles on the path.

- **Factors Influencing Fading**

The following physical factors influence small-scale fading in the radio propagation channel:

1. **Multipath propagation**-Multipath is the propagation phenomenon that results in radio signals reaching the receiving antenna by two or more paths. The effects of multipath include phase shifting, constructive and destructive interference.
2. **Speed of the mobile**-The relative motion between the base station and the mobile results in random frequency modulation due to different Doppler shifts of each MPC.
3. **Speed of surrounding objects**-If objects in the radio channel are in motion, they induce a time varying Doppler shift on multipath components. If the surrounding objects move at a greater rate than the mobile, then this effect dominates fading.
4. **Transmission Bandwidth of the signal**-If the transmitted radio signal bandwidth is greater than the bandwidth of the multipath channel i.e. quantified by coherence bandwidth then received signal will be distorted.

- **Direct wave path, line of sight path and obstructive path**

In mobile communication air is used as medium, hence signal may serve many possible paths to reach at receiver. A direct wave path will be present between transmitter and receiver when the surface of earth is plan/flat i.e. no terrain contour. If the path between transmitter and receiver having no buildings/ obstacle this condition is known as line of sight path. But practically we always do not have the line of sight or direct path. The average received signal power is grater in case of line of sight condition as compared to multipath.

- **Noise level in cellular frequency band**

There are two types of man-made noise present in cellular frequency band as follows

1. **The Ignition Noise.** The 800 MHz frequency band was not widely used in past. The ignition noise generated by the vehicle is the man-made noise at 800 MHz. The automotive noise introduced at 800 MHz by ignition noise with a bandwidth of 30 kHz.
2. **The 800-MHz-Emission Noise.** As the cellular mobile systems is

operating in all the major cities. The 800-MHz-emission noise can be measured at an idle channel. We can calculate it at a forward voice channel in from the 869 to 894 MHz region when the mobile receiver is in no traffic spot of a city. In this case automotive ignition noise will not be involved only the noise introduced by the emission will present.

1.6 OPERATION OF CELLULAR SYSTEMS

The operation of a cellular system can be divided into following parts:

- **Mobile unit initialization:**

When subscriber activates the mobile unit the receiver scans 21 set-up channels among the 416 channels. The strongest channel is selected and locks on for a certain time. Generally the strongest set-up channel selected from nearest cell site. This scheme is known as self-location scheme.

- **Mobile originated call:**

The mobile subscriber entered the called number into an originating register in the mobile unit, checks to that the number is correct or not and press the send/call button. A request for service is sent on a selected set-up channel obtained from a self-location scheme. The cell site receives it, and in directional cell sites, selects the best directive antenna for the voice channel to use. The cell site also sends a request to the MTSO through a high-speed data line at the same time. The MTSO selects an appropriate voice channel for the call, and the cell site used the best directive antenna to link the mobile unit. The MTSO also connects the wired line through the zone office of Telephone Company.

- **Network originated call:**

A land-line party dials a mobile unit number. The telephone company zone office identified that the number is mobile and forwards the call to the MTSO. The MTSO sends a paging message to some cell sites based on the mobile unit number and the search algorithm. Each cell site transmits the page on its own set-up channel. The mobile unit recognizes its own identification on a strong set-up channel, locks onto it, and responds to the cell site. The mobile unit also follows the instruction to tune to an assigned voice channel and initiate user alert.

- **Call termination:**

When the mobile user turns off the transmitter, a particular signal by a signaling tone transmits to the cell site, and both sides free the voice channel.

- **Handoff procedure:**

When the mobile unit moves out of the coverage area of a particular cell site, the reception becomes weak. The present cell site requests a handoff. The system switches the call to a new frequency channel in a new cell site without interrupting the ongoing call or alerting the user. The call continues as long as the user is talking.